Diffusion Models:
From Adding Noise to Generating Images

Understanding denoising-based generation through DDPM and LDM

Cao Quang Hung (24125010) Nguyen Trong Van Viet (24125023) Huynh Chi Ton (24125020)
University of Science, VNU-HCM · CS420 – Artificial Intelligence

GUIDING QUESTION

How can learning to remove noise enable a model to generate new images?

forward noising: $x_0 \rightarrow x_T$



$t = 0$ $t = 100$ $t = 300$ $t = 1000$

1 Forward noising
DDPM · Ho et al., 2020

2 Reverse denoising
DDPM · noise prediction

3 Latent diffusion
LDM · Rombach et al.

4 Evidence & limits
our pretrained-model runs

Learning objective: explain forward noising and reverse denoising, including why latent diffusion reduces computational demands.

Where diffusion fits: three ways to generate images

Generative vs. discriminative

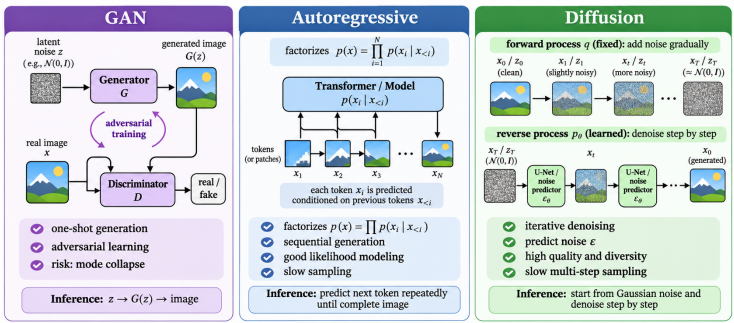
Discriminative learns $p(y \mid x)$: "is this a cat?" needs only class-separating features.

Generative learns $p(x)$ or $p(x \mid y)$: must capture shape, texture, lighting and layout.

Timeline

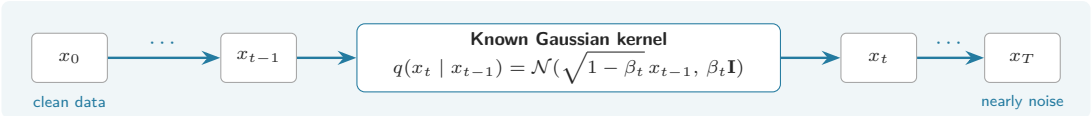
2015 diffusion framework (Sohl-Dickstein)
2019 score-based generation (Song & Ermon)
2020 DDPM: predict the noise (Ho)
2022 LDM: latent space + cross-attention (Rombach)

GAN vs. autoregressive vs. diffusion



Project illustration; sources: Goodfellow et al. 2014, van den Oord et al. 2016, Ho et al. 2020. Typical behaviour, not universal guarantees.

Forward noising: a known way to destroy structure



One step, and the jump to any t

$$x_t = \sqrt{1-\beta_t}\, x_{t-1} + \sqrt{\beta_t}\, \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, \mathbf{I})$$

$$\boxed{x_t = \sqrt{\bar{\alpha}_t}\, x_0 + \sqrt{1-\bar{\alpha}_t}\, \varepsilon}$$

$\beta_t \in (0,1)$ variance schedule, $\alpha_t = 1 - \beta_t$, $\bar{\alpha}_t = \prod_{j \le t} \alpha_j$. The index t counts steps, not physical time.

- β_t both *shrinks* the old signal ($\sqrt{1-\beta_t}$) and *sets* the new noise level ($\sqrt{\beta_t}$).
- $\mathcal{N}(0, \mathbf{I})$ is stationary, so after T steps $x_T \approx \mathcal{N}(0, \mathbf{I})$: we know where generation will start.
- DDPM schedule: $T = 1000$, β_t linear from 10^{-4} to 0.02.

Why $T = 1000$ small steps instead of one big step? **(1)** Small steps keep each *reverse* step easy: nearly Gaussian, a small correction. **(2)** Every noise level becomes a training task, from fine detail (small t) to coarse layout (large t) *(our reading)*. Training does not pay for T steps (closed form); the cost appears at sampling → slide 6.

Sohl-Dickstein et al. 2015; Ho et al. 2020 (notation α_t, $\bar{\alpha}_t$). Diagram: original schematic.

Forward noising by hand and by eye

1-D calculation OUR EXAMPLE

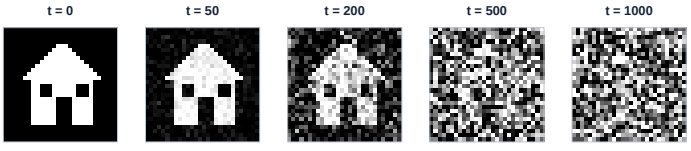
$$x_0 = 0.8,\ \beta_1 = 0.1,\ \varepsilon = -0.5 \ \Rightarrow\ \alpha_1 = 0.9$$

$$\sqrt{\alpha_1}\, x_0 = \sqrt{0.9}\,(0.8) = 0.758947$$

$$\sqrt{\beta_1}\, \varepsilon = \sqrt{0.1}\,(-0.5) = -0.158114$$

$$x_1 = \boxed{0.600833}$$

- $q(x_1 \mid x_0{=}0.8) = \mathcal{N}(0.758947,\ 0.1)$: mean $\sqrt{0.9}\, x_0$, standard deviation $\sqrt{0.1} = 0.316$.
- $\mathbb{E}[x_1 \mid x_0] = \sqrt{0.9}\, x_0$ since $\mathbb{E}[\varepsilon] = 0$; the sample 0.6008 differs because ε is one random draw.
- $\beta_1 = 0.1$ only makes the arithmetic visible; the image below uses the small DDPM schedule.

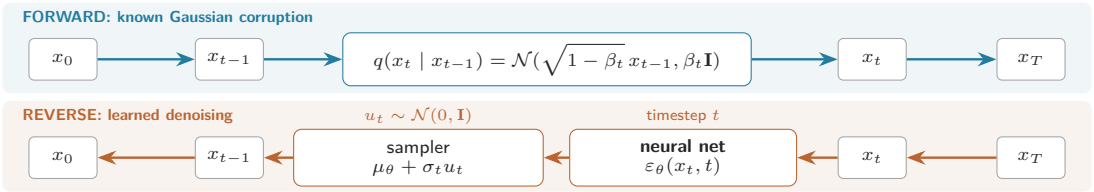


OUR EXAMPLE
28×28
house

$\mathrm{SNR} = \bar{\alpha}_t/(1-\bar{\alpha}_t)$: the house is clearly visible at $t = 50$, noise dominates at $t = 500$, almost nothing is left at $t = T$.

Our example (no model): synthetic house, $T = 1000$, linear β_t from 10^{-4} to 0.02, one sampled trajectory.

Reverse denoising: learn to undo one small step



One reverse step

$$p_\theta(x_{t-1} \mid x_t) = \mathcal{N}\Big(\mu_\theta(x_t, t), \sigma_t^2 \mathbf{I}\Big)$$

$$\mu_\theta = \frac{1}{\sqrt{\alpha_t}}\Big(x_t - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}}\, \boldsymbol{\varepsilon_\theta(x_t, t)}\Big)$$

Why predict ε, not x_0? $\varepsilon \sim \mathcal{N}(0, \mathbf{I})$ is a simple, normalized target; it also encodes the score, $-\varepsilon_\theta / \sqrt{1-\bar{\alpha}_t} \approx \nabla \log q_t(x_t)$.

Training objective

$$L_{\text{simple}} = \mathbb{E}_{t, x_0, \varepsilon}\Big[\|\varepsilon - \varepsilon_\theta(x_t, t)\|^2\Big]$$

1. draw image x_0, $t \sim \mathrm{U}\{1..T\}$, $\varepsilon \sim \mathcal{N}(0, \mathbf{I})$
2. build $x_t = \sqrt{\bar{\alpha}_t}\, x_0 + \sqrt{1-\bar{\alpha}_t}\, \varepsilon$
3. predict $\varepsilon_\theta(x_t, t)$; gradient step on the error

PAPER REPORTS CIFAR-10 FID 3.17 vs. 13.51 (variational)

Diagram: original schematic after Sohl-Dickstein et al. 2015 and Ho et al. 2020. L_{simple} is a reweighted surrogate of the variational bound (Ho et al. 2020, Tables 1–2).

From denoising to generation: answering the guiding question

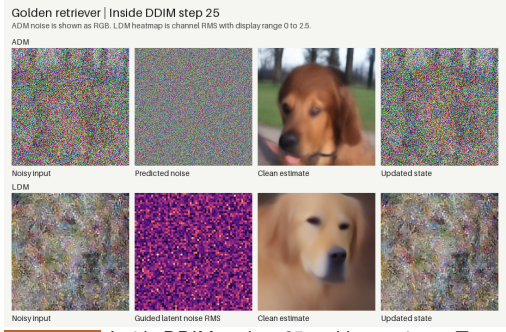
Sampling: no real image needed

1. $x_T \sim \mathcal{N}(0, \mathbf{I})$ (fresh noise)
2. for $t = T, \ldots, 1$: $x_{t-1} = \mu_\theta(x_t, t) + \sigma_t\, u_t,\ u_t \sim \mathcal{N}(0, \mathbf{I})$
3. return x_0 (no noise is added at $t = 1$)

Why removing noise generates new images

1. **Free supervision:** the forward process gives (x_t, ε) pairs at every noise level.
2. **Shared structure:** to predict the noise, the network must learn what images have in common.
3. **Small stochastic steps:** each step is a gentle correction; fresh u_t makes every seed a different image.

Why not one step? Training $x_T \mapsto x_0$ with squared error gives $\mathbb{E}[x_0 \mid x_T] \approx \mathbb{E}[x_0]$ because x_T carries almost no information about x_0: the average, blurry image. Small *forward* steps are what make small *reverse* steps possible.

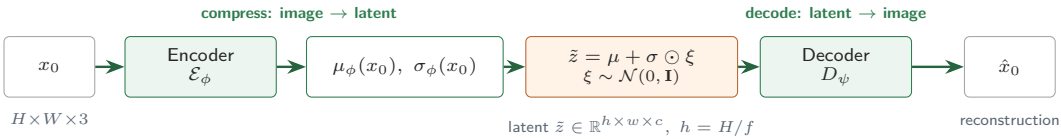


OUR RUN *Inside DDIM update 25, golden retriever.* Top: pixel ADM; bottom: latent LDM.

- The *clean estimate* already shows head and muzzle, while the *updated state* is still noisy.
- A forward pass from the final image makes the retriever's features vanish gradually (cf. slide 4).

Our run: OpenAI ADM (pixel), CompVis cin256-v2 (latent), ImageNet, 50 DDIM steps; one sample per class. Sampling: Ho et al. 2020, Alg. 2.

Inside the autoencoder: how images become latents



VAE objective

Reconstruction + KL to the prior:

$$\mathbb{E}[-\log p_\psi(x_0 \mid \tilde{z})]$$

$$+ \mathrm{D}_{\mathrm{KL}}\left(q_\phi \| \mathcal{N}(0, \mathbf{I})\right)$$

The decoder must rebuild the image; the KL keeps codes near $\mathcal{N}(0, \mathbf{I})$. **Problem:** pixel-wise squared error averages plausible details, so images turn *blurry*.

LDM: a perceptual compressor

1. *Perceptual loss*: compare deep features, so the result must *look* like the input.

2. *Patch-based adversarial loss*: a discriminator judges local patches ⇒ realistic texture.

3. *Light regularizer*: weak KL (**KL-reg**) or a codebook in the decoder (**VQ-reg**).

Ours: Churches = KL, f=8; ImageNet = VQ, f=4.

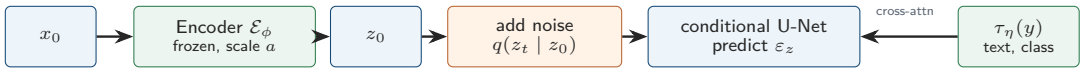
What changes for diffusion

- Latent keeps its 2-D layout, so a U-Net can process it.
- Autoencoder: trained once, then **frozen**.
- Diffusion learns the *distribution of latents*; clean latents are not $\mathcal{N}(0, \mathbf{I})$, so decoding $\tilde{z} \sim \mathcal{N}(0, \mathbf{I})$ is not enough.
- Encoder noise $\xi \neq$ diffusion noise ε_z.

Kingma & Welling 2014; Rombach et al. 2022, Sec. 3.1; Zhang et al. 2018; Isola et al. 2017; Esser et al. 2021; van den Oord et al. 2017.

Why latent diffusion is cheaper: same loop, smaller state

TRAINING: encode a real image, learn noise prediction in latent space



GENERATION: sample a latent, denoise repeatedly, decode once



Smaller state per denoising step

- $f{=}4$: $256{\times}256{\times}3 \to 64{\times}64{\times}3$, i.e. **16× fewer spatial positions**. PAPER REPORTS Table 8
- LSUN Churches (KL-8): $196{,}608$ vs. $4{,}096$ values, $\approx$**48× fewer**. OUR EXAMPLE (arithmetic from run settings)

Same principle, new space

$$L_{\text{LDM}} = \mathbb{E}\left[\|\varepsilon_z - \varepsilon_\theta(z_t, t, \tau_\eta(y))\|^2\right]$$

Forward noising and ε-prediction are unchanged; DDIM adds fewer steps on top. **Caveat:** compression caps quality (fine text, small objects), and runtime/memory depend on more than state size → slide 10.

Pipeline: original schematic after Rombach et al. 2022, Fig. 3 and Secs. 3.1–3.3; sampler: Song et al. 2021 (DDIM).

Cross-attention: steering the denoiser with a prompt

Prompt → tokens → attention

$C = \tau_\eta(y)$ (frozen CLIP text encoder in Stable Diffusion v1.4)

$$Q = FW_Q, \quad K = CW_K, \quad V = CW_V$$

$$O = \mathrm{softmax}\left(QK^\top/\sqrt{d}\right)V$$

Queries come from the image; keys and values come from the condition. A region turning into a bus can draw on the tokens *red* and *bus*.

- Recomputed at every layer and step: the prompt steers each denoising prediction; it is *not* pasted onto the final image.
- Only τ_η changes with the modality: text, class label (1 token, our ImageNet LDM), layout; maps and masks are concatenated with z_t.
- Classifier-free guidance $\varepsilon_u + s(\varepsilon_c - \varepsilon_u)$: larger s, closer to the prompt, less diverse.

OUR RUN **Stable Diffusion v1.4, seed 2027, same initial latent**



Reference
"a red bus to the left of a blue car"

red → green
"a red bus to the left of a blue car"

left → right
"a red bus to the right of a blue car"

- Colour edit: green bus at **all 4 seeds** ✓, but no clear blue car; seed 2027 shows two green buses ✗.
- Side edit: no clear left-to-right swap; the red bus stays on the left ✗.
- Cross-attention is *soft*: attributes can bind to the wrong object, or a token can be ignored.

Vaswani et al. 2017; Rombach et al. 2022, Sec. 3.3; Ho & Salimans 2022; Radford et al. 2021; Chefer et al. 2023. Images: our run, CFG 7.5, 50 DDIM steps; 16 images checked by eye, an illustration and not a general conclusion.

Pixel vs. latent diffusion on LSUN Churches

OUR RUN Pretrained checkpoints (no training by us) · Tesla T4 · batch 1 · 50 DDIM steps ($\eta = 0$) · 4 seeds (2026–2029) · warm-up excluded, GPU-synchronized timing

Criterion	Pixel DDPM	LDM-KL-8
Denoising state	$256\times256\times3$	$32\times32\times4$
Median latency (s/img)	6.10	**2.44** ($2.5\times$ faster)
Peak CUDA alloc. (GiB)	**0.75**	1.68

PAPER REPORTS Rombach et al., Table 6 (Places inpainting, 256^2): pixel LDM-1 vs. latent LDM-4 = $0.26 \rightarrow 0.97$ samples/s; FID@2k $24.74 \rightarrow 15.21$.



Pixel: clear steeple (2028), dark 2029 hides detail. Latent: clear facades (2027–2029), 2026 is a distant complex. Each model draws noise in its own state space.

Takeaway: the latent model was faster but used *more* peak memory in our setup, so "cheaper compute" is not "cheaper in every resource". Four images per model, one checkpoint pair, no FID: an illustration, not a benchmark.

Our run: ddpm-ema-church-256 vs. CompVis LDM-KL-8 (FP32); latency includes the decoder. Paper values: Rombach et al. 2022, Table 6.

Limitations: training data, copyright and bias

1. Training data

A model only knows what its data contains.

- Church checkpoints specialize in churches.
- ImageNet conditioning uses a fixed class vocabulary.
- Text models inherit the attributes and relations present in captions.

2. Bias

Bianchi et al. (2023): ordinary prompts to text-to-image systems can yield **demographic stereotypes** at scale.

Our bus-and-car demo does not measure bias.

3. Copyright and memorization

- Carlini et al. (2023) extracted **memorized training images** from diffusion models.
- Data provenance matters for copyright; the U.S. Copyright Office (2025) analyses training under *U.S. law*, not Vietnamese law.
- This does not mean every generated image is a copy.

Method limits: many sampling steps · quality capped by autoencoder reconstruction · soft cross-attention (slide 9) · our experiments are small illustrations (no FID, no bias metrics), not benchmarks.

Bianchi et al. 2023; Carlini et al. 2023; U.S. Copyright Office 2025, Copyright and AI, Part 3 (pre-publication version).

TAKE-HOME

Conclusion: answering the guiding question

How can learning to remove noise enable a model to generate new images?

Because the forward process gives **free training pairs at every noise level**, a network trained to predict the noise learns **shared image structure**, and running it from **fresh noise in many small stochastic steps** turns noise into a new sample.

1. Forward noising

Known Gaussian chain with schedule β_t; the closed form $x_t = \sqrt{\bar{\alpha}_t}\, x_0 + \sqrt{1-\bar{\alpha}_t}\, \varepsilon$ reaches any t directly.

2. Reverse denoising

Learn $p_\theta(x_{t-1} \mid x_t)$ by predicting the noise (L_{simple}); sample by iterating small steps from $\mathcal{N}(0, \mathbf{I})$.

3. Latent diffusion

Same principle on autoencoder latents: cheaper steps ($\approx 2.5\times$ faster in our run; $0.26 \rightarrow 0.97$ samples/s in the paper) and prompts via cross-attention. Costs: reconstruction limit, memory not necessarily lower.

Capability is bounded by the training data, and by what we actually measured.

Thank you · Questions?

References and figure sources: backup slides

References (1/2)

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Figures and results. The schematics (slides 3, 5, 7, 8) are original drawings of the cited methods. The 1-D calculation, the 28×28 noising sequence, the LSUN Churches timing, the ImageNet denoising traces and the Stable Diffusion prompt test are our own runs with pretrained checkpoints (no training by us); published numbers are attributed to Ho et al. 2020 and Rombach et al. 2022.

AI assistance. OpenAI Codex (report drafting, code, checks); Claude (slide drafting and Beamer preparation).